Representing piecewise functions

Recently we looked at how to use Fourier's formula to represent the function $f(x) = x$ as a Fourier series.

But we can also use a Fourier series to represent a piecewise function, and doing so will be important as we move forward.

Remember that piecewise functions are, not surprisingly, defined "in pieces." We use one expression to define the function on a particular interval, another expression to define the function on a different interval, etc. The general form of a piecewise function is

$$f(x) = \begin{cases} g(x) & x_0 \leq x < x_1 \\ h(x) & x_1 \leq x < x_2 \\ \vdots & \\ n(x) & x_{n-1} \leq x < x_n \end{cases}$$

Any function that's defined in two or more pieces is a piecewise-defined function.

So let's do another example where we find the Fourier series representation of a piecewise-defined function. The process will look exactly the same as what we're used to doing already, except that we'll split the integrals we use to find A_0, A_n, and B_n into multiple integrals, one for each piece of the piecewise function.

Example

© Krista King Math



Find the Fourier series representation of the piecewise function on $-L \leq x \leq L$.

$$f(x) = \begin{cases} 1 & -L \leq x < 0 \\ x & 0 \leq x \leq L \end{cases}$$

For A_0, we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\, dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{0} 1\, dx + \frac{1}{2L}\int_{0}^{L} x\, dx$$

$$A_0 = \frac{x}{2L}\Big|_{-L}^{0} + \frac{x^2}{4L}\Big|_{0}^{L}$$

$$A_0 = \frac{0}{2L} - \frac{-L}{2L} + \frac{L^2}{4L} - \frac{0^2}{4L}$$

$$A_0 = \frac{1}{2} + \frac{L}{4}$$

$$A_0 = \frac{L+2}{4}$$

For A_n, we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\, dx$$

© Krista King Math

$$A_n = \frac{1}{L}\int_{-L}^{0} \cos\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\left(\frac{L}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{0} + \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \left(\frac{1}{n\pi}\right)\sin\left(\frac{n\pi(0)}{L}\right) - \left(\frac{1}{n\pi}\right)\sin\left(\frac{n\pi(-L)}{L}\right) + \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = -\left(\frac{1}{n\pi}\right)\sin(-n\pi) + \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Since $\sin(-x) = -\sin x$, we get

$$A_n = \left(\frac{1}{n\pi}\right)\sin(n\pi) + \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{0}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$A_n = \frac{1}{L}\left[x\left(\frac{L}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right) - \int\left(\frac{L}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_{0}^{L}$$

$$A_n = \frac{1}{L}\left[\left(\frac{Lx}{n\pi}\right)\sin\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi}\right)^2\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_{0}^{L}$$

$$A_n = \frac{1}{L}\left[\left(\frac{L^2}{n\pi}\right)\sin(n\pi) + \left(\frac{L}{n\pi}\right)^2\cos(n\pi) - \left(\frac{L}{n\pi}\right)^2\cos(0)\right]$$

© Krista King Math

$$A_n = \frac{L}{(n\pi)^2}[(n\pi)\sin(n\pi) + \cos(n\pi) - 1]$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for A_n simplifies to

$$A_n = \frac{L}{(n\pi)^2}[0 + (-1)^n - 1]$$

$$A_n = \frac{(-1)^n L - L}{(n\pi)^2}$$

And for B_n, we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{0} \sin\left(\frac{n\pi x}{L}\right)\,dx + \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{0} + \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = -\frac{1}{n\pi} + \frac{1}{n\pi}\cos(-n\pi) + \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Since $\cos(-x) = \cos x$, we get

$$B_n = -\frac{1}{n\pi} + \frac{1}{n\pi}\cos(n\pi) + \frac{1}{L}\int_{0}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

For $n = 1, 2, 3, \ldots$, $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

© Krista King Math

$$B_n = \frac{(-1)^n - 1}{n\pi} + \frac{1}{L}\int_0^L x \sin\left(\frac{n\pi x}{L}\right)\, dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{(-1)^n - 1}{n\pi} + \frac{1}{L}\left[-\left(\frac{Lx}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) - \int\left(-\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\, dx\right]\Bigg|_0^L$$

$$B_n = \frac{(-1)^n - 1}{n\pi} + \frac{1}{L}\left[-\left(\frac{Lx}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) + \left(\frac{L}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \frac{(-1)^n - 1}{n\pi} + \frac{1}{L}\left[-\left(\frac{L^2}{n\pi}\right)\cos(n\pi) + \left(\frac{L}{n\pi}\right)^2 \sin(n\pi)\right]$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = \frac{(-1)^n - 1}{n\pi} - \frac{L}{n\pi}\cos(n\pi)$$

$$B_n = \frac{(-1)^n - (-1)^n L - 1}{n\pi}$$

$$B_n = \frac{(-1)^n(1 - L) - 1}{n\pi}$$

Then the Fourier series representation of the piecewise function on $-L \le x \le L$ is

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

© Krista King Math

$$f(x) = \frac{L+2}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n L - L}{(n\pi)^2} \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} \frac{(-1)^n (1-L) - 1}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

© Krista King Math